ROLL NO :— 04

Assigment :- Organic Chemistry

2nd Semester.

Solvent effect:-

Change in the reactivity or properties of a solute caused by presence of a solvent. Solvents can impact various aspects of solute behaviour including:-

Reactivity :- Solvents can alter the rate or outcome of a chemical reaction.
Solubility :- Solvents can effect the ability of a solute to dissolve.
Stability :- Solvent can affect the stability of solute, potentially leading to degradation or decomposition.
Spectroscopic properties:- Solvents can shift or boarden spectroscopic signals (eg NMR, IR)
Chemical equilibrium:- Solvents can alter the equlibrium position of a chemical reaction.

Solvent can be attributed to various factors=

Polarity:-
Solvents can be polar or non-polar, influencing the solute's behaviour.

Dielectric constant:-
The Solvents ability to dissolve ions and polar

molecules.

Hydrogen bonding:-
Solvents can form hydrogen bonds with solute impacting it's behaviour.

Steric effect:-
Solvents can influence the spatial arrangment of molecules.

Solvent effect on the acidity of alcohols can be explained in following ways:-

i- **In aqeous solution:-** Alcohols are neutral and do not change pH of solution.

Example:-
for example.
Methanol and Ethanol are neutral in water, dose not change the pH.

equation:-

$CH_3-OH + H_2O \rightleftharpoons CH_3-OH + H_2O$ (no reaction)

$CH_3CH_2OH + H_2O \rightleftharpoons CH_3CH_2OH + H_2O$ (no reaction)

ii **In organic solvents** the acidity of alcohols can be enhanced, allowing them to be fully deprotonated by strong bases like Sodium metal.

Sodium metal (Na) can deprotonate ethanol in tetrahydrofuran (THF) to form a sodium ethoxide.

$$2CH_3CH_2OH + 2Na \longrightarrow 2CH_3CH_2ONa + H_2 (gas)$$

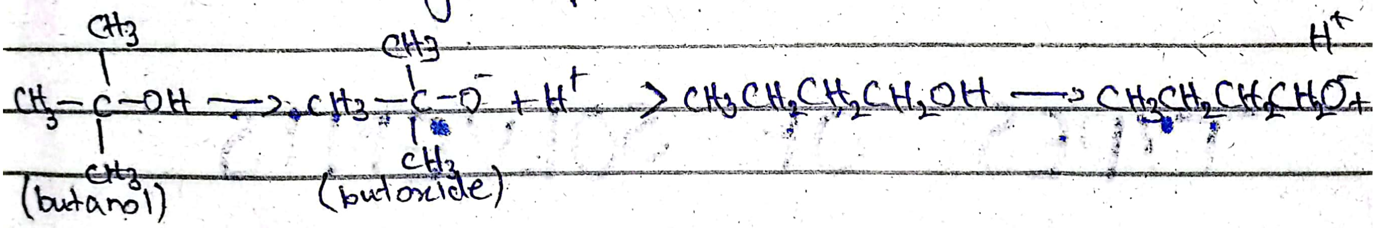
Sodium hydride can deprotonate ethanol isopropyl alcohol in dimethyl sulfoxide (DMSO) to form a sodium isopropoxide.

$$(CH_3)_2CHOH + NaH \longrightarrow (CH_3)_2CHONa + H_2 (gas)$$

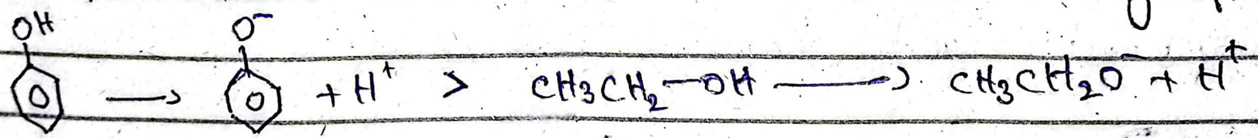
iii- **The acidity of alcohols in gas phase** is reversed compared to in solution, with tertiary alcohols being more acidic than primary alcohols.

Example:-

1- tertiary butanol is more acidic than primary butanol in gas phase.



2- Phenol is more acidic than ethanol in gas phase



iv **Solvation effect** play a crucial role in determining the acidity of alcohols with larger alkoxide ions being less well solvated.

Lithium alkoxide are more stable in solvents like THF or diethyl ether due to coordination with metal ions.

$$CH_3CH_2OLi + THF \longrightarrow [CH_3CH_2OLi(THF)x]$$

2- Potassium tert-butoxide (KOt-Bu) is more stable in DMSO than in water due to more solvation.

$KOt\text{-}Bu + DMSO \longrightarrow [KOt\text{-}Bu(DMSO)_4]$

✓ **The size of conjugate base of alcohols** influences its acidity with larger base being more stable being less acidic.

1- Larger alkoxide ions like potassium ter-butoxide are less acidic than smaller ones like potassium ethoxid.

2- Sodium phenoxide ($NaOC_6H_5$) is less acidic than sodium methoxide due to larger ~~sied~~ size of sodium phenoxide ion.

TYPES OF SOLVENTS

Solvents are broadly classified on bases of polarity and chemical nature.

Based on polarity:-

Polar solvents.

Non-polar solvents.

- **Polar solvents** have high dielectric constant and dissolve ionic and polar compounds.
- **Non-polar solvents** are better at dissolving non-polar compound.

Based on proticity:-

- **Protic solvents** can donate a proton and often participate in hydrogen bonding. They can stablize a conjugate base due to hydrogen bonding which results in increasing acidity of alcohols.
  ethanol is more acidic in water as compared to be in non-polar solvent.

- **Aprotic solvents** can not donate a proton they stablize ion through dipole interaction they increase the stability of alcohol by stablizing alkoxide ion.

ACIDITY OF PRIMARY, SECONDARY AND TERTIARY ALCOHOLS

Their acidity depends upon the stability of alkoxide ion more stable the ion is the more acidic the alcohol will be.

Primary alcohols:-

Alkoxide ion derived from a primary alcohol is least sterically hindered and can be better stablized by solvent molecule makes them more acidic.

Secondary alcohols:-

Alkoxide ion is more sterically hindered than primary reducing the stablization by solvent consequently they are less acidic.

Tertiary alcohols:-

The alkoxide ions are highly hindered, making it difficult for alcohols to stablize making them least acidic.

So the order of acidity is

Primary > Secondary > tertiary

CONCLUSION:-

Solvent effect as well as the nature of alcohol plays a very important role in determinig acidity of alcohols which depends upon the stablization of alkoxide ion in a solution.